- **Prob. 1.** Calculate the energy in joule per mole for radiations of wavelength 1000Å.
- Sol.
- Energy per quantum, $E = h\nu = hc/\lambda$
- Energy per mole or per einstein , $E = N h\nu = Nhc/\lambda$
- Here, $h = 6.62 \times 10^{-27}$ erg s, $c = 3 \times 10^{10}$ cm/s, $N = 6.023 \times 10^{23}$,
  $\lambda = 1000Å = 1000 \times ^{10-8}$ cm
- Therefore,
- $E = (6.023 \times 10^{23} \times 6.62 \times 10^{-27} \times 3 \times 10^{10}) / 1000 \times ^{10-8}$
- $= 11.94 \times 10^{12}$ ergs/mole
  $= 11.94 \times 10^{12} / 10^{7}$ J/mole
  $= 11.94 \times 10^{5}$ J/mole
  $= 11.94 \times 10^{12} / 4.184$ cal/mole
  $= 2.859 \times 10^{5}$ cal/mole

- 2. 1.0×10^{-5} mole of product is formed on the absorption of radiation of 6.62×10^{7} ergs at 3600Å. Calculate the quantum yield.

- Sol

- Number of molecules reacting = $1.0 \times 10^{-5} \times 6.023 \times 10^{23}$

  $= 6.023 \times 10^{18}$

- Number of quanta absorbed = $\frac{\text{Total energy absorbed}}{\text{Energy of one quantum}} =$

  $\frac{6.62 \times 10^{7} \text{ ergs}}{h\nu} = \frac{6.62 \times 10^{7} \lambda}{hc} = \frac{6.62 \times 10^{7} \times 3600 \times 10^{-8}}{6.62 \times 10^{-27} \times 3 \times 10^{10}}$

  $= 1.2 \times 10^{19}$

- Therefore,

- Quantum yield = $\frac{\text{No. of molecules reacting}}{\text{No. of quanta absorbed}} = \frac{6.023 \times 10^{18}}{1.2 \times 10^{19}}$

  $= 0.5016$

- 3. The bond energy in a molecule is 5.98×10^{12} ergs/mole. What is the longest wavelength of light capable of dissociating this molecule?
- Sol.
- Energy per mole = $Nh\nu = Nhc/\lambda$
- Here, $E = 5.98 \times 10^{12}$ ergs/mole, $h = 6.62 \times 10^{-27}$ erg s,
  $c = 3 \times 10^{10}$ cm/s, $N = 6.023 \times 10^{23}$, $\lambda = ?$
- Therefore,
- $5.98 \times 10^{12} = \dfrac{6.023 \times 10^{23} \times 6.62 \times 10^{-27} \times 3 \times 10^{10}}{\lambda}$
- $\lambda = \dfrac{6.023 \times 10^{23} \times 6.62 \times 10^{-27} \times 3 \times 10^{10}}{5.98 \times 10^{12}} = 2000 \times 10^{-8}$ cm
  $= 2000$Å

- 4. Radiation of wavelength 2540Å was passed through a cell containing 10 mL a solution of 0.0495M oxalic acid and a 0.01M uranyl sulphate. After the absorption of 8.81x108 ergs of radiation the conc. of oxalic acid was reduced to 0.0383M. Calculate the quantum yield.
- Sol.
- 10 mL of 0.0495M oxalic acid = 10x0.0495/1000 = 0.000495mole
- 10 mL of 0.0383M oxalic acid = 10x0.0383/1000 = 0.000383mole
- Amount of oxalic acid decomposed = 0.000495 - 0.000383
  = 0.000112 mole
- No. of molecules = $0.000112 \times 6.023 \times 10^{23} = 6.7457 \times 10^{19}$
- Number of quanta absorbed = $\frac{8.81 \times 10^{7} \text{ ergs}}{h\nu}$
- $= \frac{8.81 \times 10^{7} \lambda}{hc} = \frac{8.81 \times 10^{8} \times 2540 \times 10^{-8}}{6.62 \times 10^{-27} \times 3 \times 10^{10}} = 1.1267 \times 10^{20}$
- Therefore, quantum yield = $\frac{6.7457 \times 10^{19}}{1.1267 \times 10^{20}} = 0.5987$

- **Prob. 5.** The dissociation energy of a molecule is 102.9 kcal/mole. If the molecule is dissociated by illumination with radiation of wavelength 2537Å. What fraction of the radiant energy will be converted into kinetic energy?
- Sol.
- Dissociation energy per molecule = $102.9 \times 10^3 / 6.023 \times 10^{23}$
= 1.708×10^{-19} cal = $1.708 \times 10^{-19} \times 4.184 \times 10^7 = 7.1463 \times 10^{-12}$ ergs
- Energy of one quanta, $E = h\nu = hc/\lambda$
- Here, $h = 6.62 \times 10^{-27}$ erg s, $c = 3 \times 10^{10}$ cm/s, $N = 6.023 \times 10^{23}$,
$\lambda = 2537Å = 2537 \times^{10-8}$ cm
- Therefore,
- $E = (6.62 \times 10^{-27} \times 3 \times 10^{10}) / 2537 \times^{10-8} = 7.8139 \times 10^{-12}$ ergs/molecule
- Energy converted into KE = $7.8139 \times 10^{-12} - 7.1463 \times 10^{-12}$
- = 0.6675×10^{-12} ergs
- Fraction converted into KE = $(0.6675 \times 10^{-12} / 7.8139 \times 10^{-12}) = 0.0854$